INTRODUCTION

THE PROJECT

SENTINAL

AI-assisted visual safety monitoring

PRESENTED BY

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PROBLEM STATEMENT

Passive CCTV footage can be difficult to monitor continuously, delaying the recognition of important safety incidents.

FROM PASSIVE VIDEO TO REVIEWED INCIDENTS

SEE. ASSESS. ALERT.

- Camera intake
- Visual inference
- Operator review

Problem Statement

What problem are we solving?

Conventional CCTV primarily records events. Detecting a developing incident still depends on an operator noticing the right moment across multiple video feeds.

Who experiences it?

- Campus security teams
- Commercial facility operators
- Control-room personnel
- Public-space safety staff

Why does it matter?

- Potential events can be missed
- Continuous attention is difficult to sustain
- Manual footage review is time-consuming

THE GAP

A practical system must connect camera feeds, usable detection models, event filtering and a clear human-review workflow.

VIDEO IS DATA. AN ALERT MUST BE EVIDENCE-BASED. RESPONSE REMAINS HUMAN-LED.

SENTINAL

AI-ASSISTED CCTV MONITORING / PROPOSED SOLUTION

HOW THE SYSTEM WORKS

01	**Connect**	Select a Mac, iPhone or supported IP camera
02	**Interpret**	Process frames with trained vision models
03	**Correlate**	Check detections across time
04	**Notify**	Generate reviewable warnings
05	**Resolve**	Record the operator's decision

HOW IT ADDRESSES THE PROBLEM

- Prioritizes candidate incidents for review
- Provides frame-level visual evidence
- Creates an auditable incident trail

DEVELOPMENT REALITY

V0.1.2 / REPORTED

Camera selection, ONVIF discovery, video import and test infrastructure.

V0.2 / INTEGRATION

Weapon, fire/smoke detection and incident alerts.

Technology & Implementation

UNIFIED TECHNICAL PIPELINE

01

VIDEO SOURCE

Mac / iPhone
Wi-Fi IP CCTV
Imported footage

→

02

INGESTION

ONVIF discovery
RTSP / decoding
Frame sampling

→

03

VISION AI

OpenCV + PyTorch
YOLO model candidates
Boxes / confidence

→

04

EVENT ENGINE

Temporal checks
Deduplication
Incident metadata

→

05

REVIEW UI

Visual alerts
Evidence preview
Operator decision

REPORTED IN PROTOTYPE V0.1.2

- Explicit camera selection; ONVIF discovery
- 261 automated tests passed (build report)

V0.2 INTEGRATION PRIORITIES

- Dedicated weapon / smoke / flame models
- Temporal assessment and reviewable alerts

Note: specific framework choices and model benchmarks must be checked against the latest repository.

Impact, Innovation & Feasibility

01 / DIFFERENTIATION

Beyond recording

- Detection + event assessment
- Human-reviewable evidence
- Modular camera connections

02 / TARGET USERS

Safety operators

- Campus security teams
- Offices and shared facilities
- Control-room personnel

03 / INTENDED IMPACT

Faster awareness

- Surface important footage
- Make alerts easier to assess
- Keep a record of decisions

04 / SCALABILITY

Camera to network

- Reusable source adapters
- Edge processing where practical
- Central incident coordination

05 / APPLICATIONS

Real-world settings

- Educational institutions
- Workplaces and facilities
- Authorized public-space monitoring

06 / FEASIBILITY

Test before trust

- False alarms need measurement
- Real CCTV support needs testing
- Privacy and oversight are essential

Expected benefits are design goals. No field-deployment impact or accuracy figures are claimed.

Reference / Demo / Closing

HACKATHON DEMONSTRATION FLOW

01 **Select a source**

Mac camera or a lawful prerecorded test clip

02 **Run inference**

Draw bounding boxes from the real model output

03 **Evaluate event**

Check confidence and temporal consistency

04 **Review an alert**

Inspect evidence, acknowledge or dismiss

SENTINAL / P.Bhupesh / THANK YOU

PROJECT STATUS

V0.1.2 / reported built

Camera sources, ONVIF and test coverage.

V0.2 / work in progress

Validated hazard detectors and incident alerts.

TECHNICAL REFERENCES

ONVIF — Camera interoperability

OpenCV — Image processing

PyTorch — Model execution

Ultralytics — YOLO models